Route Planner

Customer integration white paper



A map-based approach to planning customer visits inside an existing business application.

> **Purpose**
>
> Describe the product workflow, architectural boundaries and implementation decisions a customer team can use to build a comparable capability in its own application.

Prepared for product owners, engineering leads and implementation teams. Neutral customer edition • Version 1.0 • October 28, 2026.

This document contains conceptual design and product behavior. Source code, passwords, access tokens, private service addresses, deployment configuration and customer records are excluded.

01 / Executive overview

A practical planning workspace for field teams

A useful route planner connects customer context, geography and the working calendar. It helps a representative decide which accounts to visit, assemble a sensible day and preserve the resulting appointments without repeatedly switching between separate tools.

The reference product supports field-sales and customer-visit planning. Its foundation is a map of permitted customer accounts, geographic selection, a draft visit sequence and a user-owned schedule. Customer information remains connected to the host application.

Capability	Reference behavior	Business purpose
Customer map	Individual account pins, customer details and assigned representatives.	Understand a territory in context.
Visit prioritization	Customer-type filters and visit-age bands, including never visited.	Identify accounts needing attention.
Area selection	Radius and freehand area selection, with individual stop removal.	Build a focused group of visits.
Day planning	Geographic stop order, manual reordering and estimated visit times.	Turn a selection into a workable draft.
Calendar workflow	Save a day, book or move one visit, replace or clear route-generated stops.	Keep the plan actionable.
Field context	Optional device location, road-route display and a traffic overlay.	Orient the user while planning.

> **Scope distinction**
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> The reference is a customer-visit planner. Multi-vehicle dispatch, load capacity, driver work rules, guaranteed arrival windows and fleet tracking require additional design.

Expected value

The intended outcomes are less manual planning, better visibility of neglected accounts and clearer coordination between planned and logged visits. Performance and business gains should be measured in a customer pilot; no quantified savings are assumed here.

02 / The user workflow

From customer discovery to a scheduled day

Step	User action	System response
1. Set the view	Choose permitted representatives, customer types and visit-age filters.	Load the relevant customer pins and retain the user’s filter preferences.
2. Explore	Pan the map, search accounts or open a customer.	Show account context; load contact details when needed.
3. Select	Draw a radius or freehand area; adjust the selected customers.	Create an explicit selection without silently changing it when filters change.
4. Build a day	Choose a date, starting point, start time and visit duration.	Suggest a geographic sequence and approximate arrival/departure times.
5. Review	Reorder or remove stops and inspect the proposed itinerary.	Keep numbered stops and the displayed road route in the same sequence.
6. Commit	Save the day, or explicitly replace a previously planned day.	Persist the selected appointments and refresh the calendar.
7. Follow through	Log completed visits in the host customer workflow.	Update the representative’s schedule and the customer’s visit history.

A simple illustrative use case

A representative filters to overdue accounts in a permitted territory, selects five nearby customers and reviews their recent activity. The planner proposes a sequence. The representative moves one appointment to accommodate a known preference, checks the estimated day and saves it.

This example is illustrative and contains no real account data. Selection and manual review remain important: the shortest-looking geographic sequence may not reflect customer availability, road access or a business priority.

> **Ownership**
>
> Reference calendar entries belong to the signed-in user. Scheduling on another person’s behalf is a separate customer-application requirement, with explicit permissions and ownership rules.

03 / Planning method and its limits

Keep draft estimates, road display and traffic separate

The reference draft planner orders valid geographic points using a nearest-neighbor sequence followed by bounded local improvement. The user can override the order. This is a practical geographic heuristic; it does not establish a mathematically optimal road route.

Planning layer	What it does	What the customer should understand
Stop sequencing	Suggests an order using geographic distance between selected accounts.	It does not solve customer time windows, road restrictions or vehicle constraints.
Draft timing	Adds estimated travel allowances and the selected visit duration.	Times are planning estimates rather than live traffic-based arrival promises.
Road-route display	Requests and draws a provider-generated driving path in the chosen order.	Drawing the road path does not automatically replace draft schedule times.
Traffic overlay	Shows traffic conditions as map context.	The reference overlay does not recalculate the stored itinerary.
Manual control	Lets the user reorder stops before committing.	Human changes should remain visible and should not be silently undone.

Selection priority and route order

Visit recency helps the user choose accounts. The reference visit-age bands distinguish never visited, under 30 days, 30–89 days and 90 days or more. Once customers are selected, geographic sequencing is a separate operation. Revenue-weighted scoring is an optional enhancement.

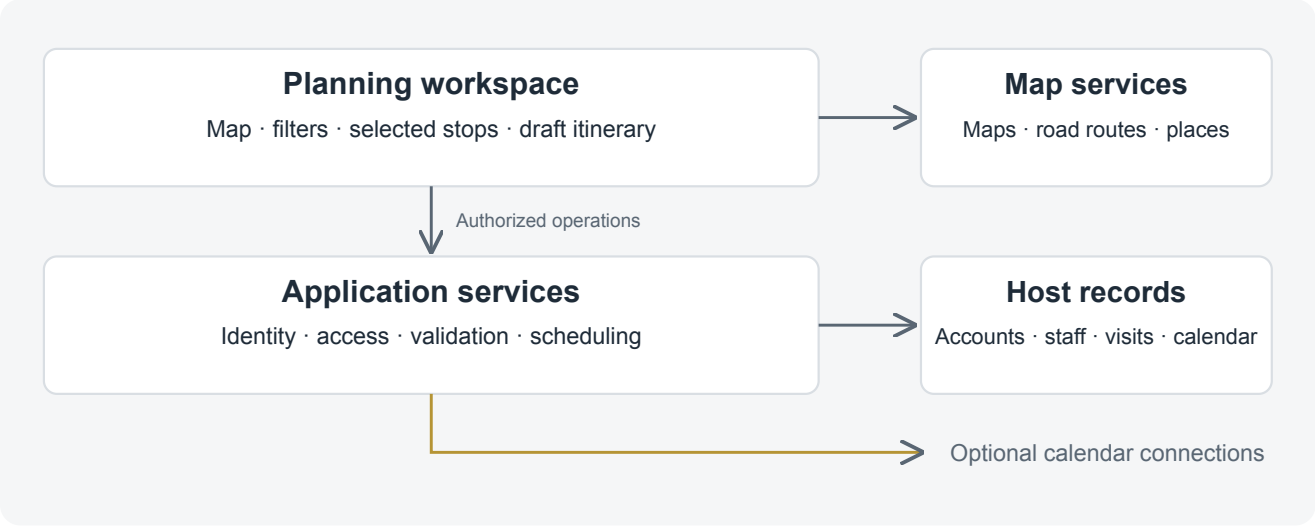
When to introduce a different planning engine

Use a road-time matrix or constraint-solving service when the target workflow needs traffic-aware timing, fixed appointment windows, breaks, multiple representatives, an explicit return trip or vehicle restrictions. Keep the scheduling model separate so the planning engine can evolve.

> **Capacity**
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> Calendar save limits and a mapping provider’s route-request limits are different concerns. The reference bounds both. A customer implementation should align its published limits, explain an oversized selection and preserve every stop when splitting provider requests.

04 / Reference architecture

Separate the interaction layer from authoritative services



Conceptual integration boundaries. The host application retains ownership of business records.

Layer	Responsibility
Presentation and draft planning	Map interactions, selected stops, filters, itinerary preview and customer panels. The reference computes draft ordering and approximate timing in the browser.
Application services	Authenticate the user, enforce permitted operations, return scoped data, validate route saves, manage calendar persistence and announce schedule changes.
Authoritative records	The host application owns customer identities, staff assignments, visits and appointments. Provider search results remain separate until explicitly saved.
External services	Map rendering, road directions, place search and optional calendar connections. These use the customer's own approved service accounts.

The reference uses React and TypeScript in the interface, Node.js and Express services, MongoDB persistence and Google Maps services. These are reference implementation choices; the customer can implement the same boundaries within its established stack.

> **Recommended adaptation**
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> Keep authorization, customer-scope checks, validation and final persistence on the backend. A customer may also centralize route generation there for consistent policy, larger plans or a different optimization service. That backend planning engine is a proposed adaptation, not a description of the current browser preview.

05 / Data and integration model

Use the host application’s identities and business records

Logical record	Essential information	Integration rule
Customer account	Stable identity, display name, address, coordinates, coordinate-quality metadata, type and status.	Reuse the host customer record; avoid a second independent directory.
Representative	Stable staff identity, assigned branches and permitted actions.	Resolve identity and scope from the authenticated session.
Visit history	Customer, representative, visit date and the recorded outcome or activity.	Use actual logged visits to calculate recency.
Draft itinerary	Selected accounts, order, origin, date, visit duration and estimated times.	Keep a preview distinct from committed appointments.
Scheduled appointment	Owner, customer, start/end, activity, status and creation source.	Distinguish route-generated entries from manually booked meetings.
User preferences	Representative filters, customer-type filters, visit-age filters and map color mode.	Store per user; apply saved preferences before the first scoped load.
Prospective lead	Provider reference, business identity, address and map point.	Convert to a customer only after explicit user action and host validation.

Location and date quality

A missing or invalid coordinate cannot form a reliable stop. Preserve geocoding provenance and confidence, and provide an address-correction workflow. A valid map point alone does not prove that a destination entrance is correct.

Define the appointment time zone, working-day boundaries, daylight-saving behavior and calendar week start before integration. Preserve the intended local date when exchanging appointments with calendar services.

Host application connections

The main touchpoints are the customer directory, staff directory, visit history, calendar and customer-editing workflow. Provide a permission-aware adapter for each. The planner should not require the customer to adopt the original application’s accounts, database or internal service addresses.

06 / Recommended backend responsibilities

A target integration contract, expressed without code

The following service responsibilities describe the recommended customer implementation. They are conceptual operations rather than published endpoint specifications.

Operation	Backend responsibility
Read the territory	Return only permitted representatives, customer counts and the requested customer pins. Exclude unnecessary contact details from the map response.
Open an account	Recheck customer access, then return the details the user is allowed to view.
Prepare a route	Validate the selection and planning options. If planning is centralized, calculate the sequence and estimates under a versioned policy.
Commit or revise a day	Validate every customer, date, owner and allowed operation. Save the entire accepted change atomically.
Book or move a visit	Check appointment ownership and apply the customer's availability and conflict policy.
Refresh clients	Announce that the relevant schedule changed; let permitted clients reload current data.

A safe route-save sequence

- Resolve the signed-in user and allowed organization, branch and customer scope.
- Validate all selected customer references, dates, time ranges and capacity rules before writing.
- Recheck the current plan when replacing or editing it; reject a stale change under the chosen concurrency policy.
- Write the complete accepted change in one transaction; preserve unrelated meetings and other users' calendars.
- Return the saved result and notify affected clients after the commit succeeds.

> **Confirmed reference behavior**
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> Whole-day creation and replacement are transactional. Replacement is separately permission-controlled and limited to the owner's route-generated entries on the chosen date. Additional controls such as idempotent retries, plan-version checks and complete write-scope revalidation are customer-integration recommendations.

07 / Security, privacy and platform readiness

Make external connections and authority explicit

Customer implementation controls

- Reuse the host identity system. Enforce allowed customer, branch and appointment access on every backend operation. Add tenant isolation when serving multiple organizations.
- Separate viewing, route creation, replacement, clearing and customer editing into explicit permissions.
- Protect confidential server credentials and calendar grants. Use application restrictions for client-facing map integrations and exclude sensitive values from logs and exports.
- Retain who changed a committed plan and when, under the customer’s audit and retention policy.

Location access

The reference asks for device location through an explicit user action when permission is needed. If access is already allowed, the location marker updates while the planner is active and visible. Location-marker movement does not repeatedly change the chosen route origin. This is foreground context, not an employee tracking or attendance system. Browser location access follows the browser permission model. [3]

Mapping service selection for a new application

The reference road overlay uses a legacy directions integration. Google’s current documentation states that new Cloud projects can no longer enable Directions API (Legacy) as of October 2026, and identifies the Routes Library as its replacement. A new customer build should select currently supported routing interfaces rather than reproduce the legacy dependency. [1, 2]

Calendar connections

The reference includes a private calendar subscription and optional provider connections. The customer should define the direction of synchronization, calendar ownership, disconnection behavior and conflict handling for each provider. Calendar subscriptions should be revocable and separate from ordinary sign-in credentials. iCalendar is a standard exchange format, not a guarantee of immediate two-way synchronization. [4]

08 / Performance, resilience and field usability

Design for realistic territories and imperfect connectivity

Concern	Reference approach	Customer integration requirement
Large territories	Only checked representatives' pins are requested; map responses are lightweight.	Measure realistic record volumes, payload size and authorization cost. Add server-side bounding-box reads or pagination when needed.
Map rendering	Individual pins outside a padded viewport are not drawn; selected route stops remain visible.	Profile dense areas and preserve individual account selection if clustering is introduced.
Account detail	Contact information loads when an account is opened.	Invalidate cached details after edits and clear inaccessible information after access changes.
Stale responses	Map loading distinguishes newer refreshes from older responses.	Cancel or ignore superseded work; never show an old route as the result of a new selection.
Calendar changes	Relevant calendar ranges reload after saves and schedule-change notifications.	Keep separate devices consistent and make refresh failures visible.
External service failure	Customer data, the draft itinerary and the map overlay have separate responsibilities.	Explain failed provider requests and permit controlled retry without duplicate appointments.

Mobile interaction

Use responsive account panels and a compact control rail. Test scrolling inside the host application, usable touch targets, keyboard behavior, long stop lists and map gestures. Area drawing needs device testing; mobile layout alone does not prove that every desktop gesture is suitable for touch.

Connectivity and operating measures

The reference depends on network access for maps, lookups and schedule writes. Offline route viewing, queued edits and conflict resolution are additional requirements. Track map and detail load times, save failures, retry duplication, provider usage and user-reported planning errors.

> **Evaluation**
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> Automated geometry, filtering and persistence tests are useful, but they do not establish provider availability, live road accuracy or performance at the customer's scale. Include a field pilot and tests against the chosen provider configuration.

09 / Implementation sequence

Deliver in stages with an acceptance gate for each

Stage	Deliverable	Acceptance gate
1. Integration discovery	Agreed users, customer scope, calendar ownership, identity adapter and map-provider selection.	Product and engineering agree on data ownership, operating limits and the initial use case.
2. Scoped customer map	Customer pins, filters, visit-age display, account details and saved preferences.	Out-of-scope records cannot be accessed; empty and missing-coordinate states are clear.
3. Draft day planning	Selection tools, manual stop control, origin, estimated itinerary and road display.	The planned, numbered and displayed stop sequences agree; estimates are clearly labeled.
4. Reliable scheduling	Single-visit booking, whole-day save/replacement, permissions and transactional persistence.	A failed write preserves the previous day; unrelated appointments remain intact.
5. Field and calendar integration	Logged-visit feedback, device-location behavior and optional calendar connections.	Dates remain correct across devices and time zones; disconnection and recovery are tested.
6. Pilot and rollout	Representative pilot, operating dashboards, support process and release criteria.	The agreed usability, data-quality, reliability and capacity targets are met.

Acceptance scenarios worth demonstrating

- A permitted representative sees only the intended territory; a direct request cannot bypass that scope.
- The same selected customers remain in the itinerary after a filter change or manual reorder.
- A failed replacement, repeated submission or concurrent edit is handled under the agreed recovery policy.
- Missing geocodes, unavailable location permission and a failed map request have understandable outcomes.
- Mobile users can reach every stop and every form action; planned dates remain stable through calendar exchange.

This sequence is an implementation plan, not a fixed delivery estimate. Effort depends on the customer’s existing identity, data quality, scheduling rules, provider account and integration test environment.

10 / Optional enhancements

Prioritize extensions against the customer’s actual workflow

Extension	When it is useful	Additional design needed
Road-time-based planning	Traffic and road geography materially affect the day.	Provider matrices, departure-time assumptions, operating limits and a replan policy.
Appointment constraints	Customers require time windows or different visit durations.	A constraint-aware planner, conflict handling and explicit exceptions.
Team planning	Managers allocate work across representatives.	Delegated authority, ownership changes, workload policy and concurrency controls.
Fleet dispatch	The use case includes deliveries or service vehicles.	Vehicle capacity, restrictions, work rules, depots and dedicated dispatch operations.
Offline field mode	Connectivity is intermittent.	Encrypted local storage policy, stale-data indicators, queued writes and conflict resolution.
Commercial prioritization	Visit value matters alongside geography and recency.	A transparent scoring policy, explainable recommendations and measured business outcomes.
Operational analytics	The customer needs evidence of planning improvement.	Agreed definitions, event collection, planned-versus-actual measures and privacy controls.

Measure the outcomes, not just the map

Useful pilot measures include time spent planning a day, coverage of overdue accounts, planned versus completed visits, changes made after route creation and scheduling failure rates. Measure actual road distance or elapsed travel from an appropriate source if those outcomes matter; do not present straight-line draft distance as actual travel.

> **Scope control**
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> These extensions are proposed capabilities. They should not be described as features already delivered by the reference planner. Choose a small initial scope, make its limitations visible and expand after the pilot demonstrates value.

11 / Customer decisions and references

Resolve the integration choices before committing the build

Decision	Questions to resolve
Primary workflow	Is the goal sales visits, service appointments or delivery dispatch? Which outcome will define success?
Identity and data scope	Who may view which customers, and who may create, replace or delegate a route? Is the application multi-organization?
Customer data readiness	Are account identities stable? Are branch, owner, visit history and destination coordinates reliable?
Scheduling rules	Which calendar owns the plan? Are durations, working hours, time windows and time zones fixed or configurable?
Mapping and operations	Which supported provider interfaces will be used? What stop limits, failure behavior and operating budgets are acceptable?
Rollout readiness	Who owns the pilot, test data, support process, performance targets and final acceptance?

Public standards and platform references

[1] Google Maps Platform — Directions Service (Legacy)
https://developers.google.com/maps/documentation/javascript/legacy/directions

[2] Google Maps Platform — Routes Library overview
https://developers.google.com/maps/documentation/javascript/routes/overview

[3] W3C — Geolocation specification
https://www.w3.org/TR/geolocation/

[4] RFC Editor — RFC 5545, iCalendar
https://www.rfc-editor.org/info/rfc5545/

Public platform references were checked on October 28, 2026. Provider capabilities and access conditions should be rechecked when the customer implementation begins.

How to use this paper

Use the reference behaviors as a functional baseline, the recommended backend responsibilities as an integration blueprint and the acceptance gates as a review checklist. Agree on the customer’s architecture and release criteria before turning the design into a delivery commitment.

> **Document boundary**
>
> This is a neutral product and architecture paper. It contains no application source code, executable examples, private addresses, passwords, access credentials or real customer records.